

THE SOLUTION /
BENCHMARK

AI Swarm Drones Find Survivors 3-4x Faster Than Traditional SAR

- 78-85% survivor detection on first pass
- 10-18 min per sq mile speed
- Mesh networking works in GPS-denied zones
- 24/7 autonomous operation for 11+ days
- False positive rate down to 12%
- Cost per deployment: \$12K vs \$15K

MARKET VALIDATION & GLOBAL SAR
ADOPTION RESEARCH

SWARM SEARCH EFFICIENCY

BENCHMARK MATRIX

Search & Rescue Performance Comparison

EMPIRICAL
BENCHMARK

METRIC	TRADITIONAL	SINGLE DRONE	AI SWARM (SUTRA)
Deployment Time	45- 90 min	15-30 min	8-15 min
Area Coverage / mi ²	2-3 hours	45-60 min	10-18 min
Detection Rate	65- 70%	55-65%	78-85%
Personnel Required	15- 25 crew	2-3 pilots	1-2 operators

SPEED & EFFICIENCY COMPARISON CHART

3-4X
FASTER
COVERAGE

Area Survey Rate 4.5 mi²/hr vs
(sq miles / hr) 0.4 mi²/hr

Traditional Ground		0.4 mi ²
Single Drone		1.2 mi ²
AI Swarm (SUTRA)		4.5 mi ²

False Positive Drop 35% →
12% (-67%)

Continuous Flight 24/7
(11+ Days)

\$8.94B

By
2033
(18.4%
CAGR)

+412%

Disaster
Surge
(20
Yrs)

37

Nations
Mandated
SAR

\$180M

FEMA
2025
Allocation